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Knowledge and strategies in diagnostic skill

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A theoretical framework for diagnosis in technical environments is presented, consisting of three layers. At the first layer, the task structure, top-level goals of the diagnostic tasks are identified that have to be fulfilled during task execution. This task structure may also be viewed as a global strategy to carry out the diagnostic task. The second layer of knowledge consists of the relevant local strategies by means of which values are obtained for goals in the task structure. The third layer consists of underlying domain knowledge. This theoretical framework is used to interpret the results as presented in the literature on diagnosis in technical environments. Finally, based on this framework, recommendations are made with respect to the training of diagnostic skill.

1. Introduction

Diagnosis is generally assumed to be a complex task to accomplish. Quite a lot of research has been carried out characterizing diagnostic behaviour and skill from different perspectives and with different interests.

To enable interpretation of data collected in diagnostic situations, it would be fruitful to develop a comprehensive framework for this purpose. This paper will discuss such a framework, consisting of several layers, followed by a discussion of results obtained in the literature interpreted in terms of these layers. Finally, implications of this type of analysis for training issues will be discussed, and conclusions about the fruitfulness of the framework will be drawn.

2. Diagnosis: a theoretical framework

To enable interpretations of collected data on diagnosis in technical environments, a theoretical framework about human behaviour in complex cognitive tasks is needed. This section will be devoted to the outline of such a framework. In the literature on performance on complex cognitive tasks, quite commonly a distinction is made between declarative and procedural knowledge (e.g., Anderson 1983, 1987). Declarative knowledge may be conceived of as a collection of stored facts, and is also called system or device knowledge in the domain of technical systems. Examples are knowledge about normal values of certain parameters, or knowledge about the function of the system. Procedural knowledge (knowledge about how-to-do-it) can be regarded as a collection of actions or procedures with which one investigates a device to make diagnoses about its disfunctioning, for example the use of the oscilloscope to test certain functions of a system. Procedural knowledge is content-specific and thus only applicable in a limited domain.

In addition to the declarative–procedural distinction, a distinction can be made between domain-specific knowledge and strategic or metacognitive knowledge. This strategic or metacognitive knowledge (knowledge about how-to-decide-what-to-do-

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and-when) is applicable across specific content domains, but remains geared towards one task (e.g., diagnosis). For example, in diagnosis, regardless of the domain, one would first *identify and interpret symptoms*, followed by *an investigation of possible reasons*, which will be *tested*, before one will apply a certain *repair or remedy*.

At a very general level, problem-solving strategies may be identified which have to do with very general thinking and reasoning skills, such as means-end analysis, reasoning by analogy, or working backwards. These general problem-solving strategies, or *weak methods* as they are sometimes called, are applicable across specific domains and across specific tasks.

With this decomposition, it is assumed that *procedural* and *device* knowledge are organized and deployed by goals, plans, and decision rules that comprise *strategic* knowledge. Thus, strategic knowledge can be said to have a *control* function to enable dynamic, flexible reasoning. As described in Gott (1989), support for the concept of strategic control knowledge comes from a number of academic domains such as geometry (Greeno 1978), text editing (Card *et al.* 1983), computer programming (Anderson *et al.* 1984), simple device operation (Kieras and Bovair 1984) and electronic troubleshooting (Gott and Pokorny 1987). The importance of strategic knowledge has been shown by Greeno (1978). He conducted an experiment on the acquisition of high school geometry knowledge showing that strategic knowledge is needed for setting goals that organize problem-solving activity. One of the students in Greeno's study knew the problem-solving operators and the geometric patterns to achieve them, but was unable to solve the problem. This result can be explained by the hypothesis that the student lacked knowledge about the problem-solving strategy needed in this problem. Schoenfeld (1979) showed that students who were given explicit instruction in the use of heuristic strategies in the domain of college mathematics, showed superior performance compared to students who had not received this training.

If the above results are put together, the following framework emerges. At the top-level, strategic knowledge is employed, consisting of several goals that have to be fulfilled during task execution. Examples of these goals in the diagnostic task may be 'Identify symptoms', 'Determine possible causes', and 'Identify repair'. As the reader may have noticed, these goals may also be viewed as various subtasks of the diagnostic task. This layer of knowledge is called knowledge about the *task structure*, or the *global strategy* (Wognum 1990). The order in which those goals are fulfilled is often flexible, and depends on the specific task-situation at hand. Each goal, however, only defines *what* intermediate conclusion has to be deduced, not *how* the intermediate conclusion has to be deduced. The knowledge about how to perform a (sub)task is called the *local strategy*, and may consist of either procedures with a fixed order (how to task a certain part), or may consist of a more flexible sequence of steps: a strategy. The local strategy of a subtask in turn selects the *domain knowledge* necessary to achieve the goal of the task. If a person becomes very experienced with the task at hand, he or she may use heuristic shortcuts in his reasoning process to obtain values for the various goals determined at the task level. For example: if that symptom occurs, it is rather likely that it is caused by that fault. The heuristics are bound to the task at hand: a person very experienced in one domain, which may turn him or her into an efficient problem-solver regarding that domain, does not necessarily show that same proficiency in other domains. On the other hand, if a person has no experience at all with the task at hand, he or she is bound to use very general reasoning methods: weak methods.

The question is what these different reasoning steps, the global strategy, the local strategies, and the underlying domain knowledge, entail in diagnosis in technical environments. The following sections will discuss data pertaining to these issues.

3. Diagnosis in technical environments: the task structure

In real-life diagnostic tasks one can distinguish between several discrete steps in the reasoning process. In a study on diagnostic reasoning of a service engineer, Jansweijer *et al.* (1989) distinguish between two main tasks the service engineer has to perform: the diagnosis task, basically concerned with generating and testing possible hypotheses about faults in the device, and the prediction task that is concerned with deriving expected behaviour of (assemblies) of components. This is a rather broad distinction though, and the question is what a finer-grained analysis of those tasks reveals.

Based on experience with the development of knowledge-based systems aimed at diagnosis, Schaafstal (1991) proposes a model consisting of a number of steps. The steps proposed show some similarity to the steps in diagnostic reasoning put forward by Wognum (1990), but are more extended, and further elaborated. The following steps may be distinguished.

3.1. *Identification of symptoms*

Symptoms are the first indication that there is something wrong. This does not necessarily have to lead to an alarm situation, but it may be just a process value that is moving towards the defined limits. 'Being tired' is not necessarily a symptom for a certain disease, but when you think that you are now more tired than you were before, you should start wondering about possible causes. 'The machine is slightly more noisy than yesterday.' In this example 'being noisy' is not the symptom, but 'being more noisy' may possibly be. The reader may notice that one needs expertise to interpret a certain signal as a symptom. Symptoms can be hard to describe, especially when they have a large perceptual component. In technical environments it happens quite often that operators or maintainers debate about the appearance of symptoms, or have very idiosyncratic names for symptoms, which makes uniform communication about them rather difficult.

Depending on the domain, symptoms may be easier or harder to detect. In industrial domains it is often quite obvious that there is a problem. On the other hand, for medical domains it may be a lot harder due to vagueness of presentation of symptoms.

3.2. *Judgement: how serious is the problem?*

Depending on a judgement about the seriousness of the problem, the whole line of reasoning and action-taking may change. If a problem is serious, it is important to take some action right away, for example to save someone's life or to prevent that any other unsafe situations will appear. It may also include a prevention of halting of the installation. When the necessary actions have been taken, the normal diagnostic routine may still take over, but that depends on the situation.

A correct judgement in this sense is very important in many domains, especially when human safety is at risk, but also in process control situations.

3.3. *Determination of possible faults*

For a certain symptom or set of symptoms there is often more than one possible underlying fault. For example for quality problems in paper and board manufacturing,

it is not unusual that there are more than thirty possible faults that may result in the symptom to diagnose. Evidence by Mehle (1980), and Gettys *et al.* (1980), cited in Morris and Rouse (1985) showed that people have difficulty in generating complete sets of hypotheses. Mechanics presented with automobile problems were generally unable to identify all of the possible causes of these problems. However, subjects were frequently over-confident about the completeness of their lists. In real-life situations, especially when 'tunnel vision' appears in situations of stress, it is likely that people 'forget' to take some of these possible faults into account. Especially with shift work it occurs quite often that not all operators are aware of all possible faults, due to the fact that they have not been exposed to all of them.

3.4. *Ordering of faults according to likelihood*

The ordering of possible faults according to likelihood is a process in which certain probabilities are 'assigned' to each possible fault. In this way, an ordering of the list of possible faults appears, with the most likely candidates on top. It should be the input to the process of testing.

3.5. *Testing*

Testing has the function of selecting the 'right' fault out of many, by ruling out as many of the other candidates as possible. Testing often takes place by collecting additional evidence, and may, especially in humans, be more like 'finding extra arguments for what you already thought it would be', than 'find ways to rule out this possibility' ('confirmation bias').

3.6. *Determination of repairs or remedies*

Repairs can take several forms. *Local* repairs apply to the exact fault at hand, and are meant to alleviate a specific problem. This is an example of *compensation for faults* (Rouse 1982, cited in Bainbridge 1984). *Global* repairs, on the other hand, seem to work in many situations, and can sometimes be applied without a full diagnosis of the problem. A local repair in the paper industry would be for example cleaning a dirty element. A more global repair would be slowing the speed of the paper machine down, since that seems to remedy many problems, but is in general not highly recommended. A global remedy in medicine may be 'staying home for a day'. This is an example of *compensation for symptoms* (Bainbridge 1984).

3.7. *Determination of the consequences of the application of repairs*

In many domains it is important to realize what consequences the application of a repair might have. For example, what side-effects do certain medications have, or in a paper mill, does application of this repair imply that the installation has to be halted. The outcome of this reasoning process may have an influence on the choice of repair, if there is more than one possibility.

3.8. *Evaluation: has the situation improved?*

The final step in diagnosis would be an evaluation of actions undertaken in terms of improvement upon the situation. Has the problem been solved or are other actions necessary?

Based on a study of expert and novice operators in the paper industry, Schaafstal (1991) concluded that the strategies used by expert and novice operators are rather

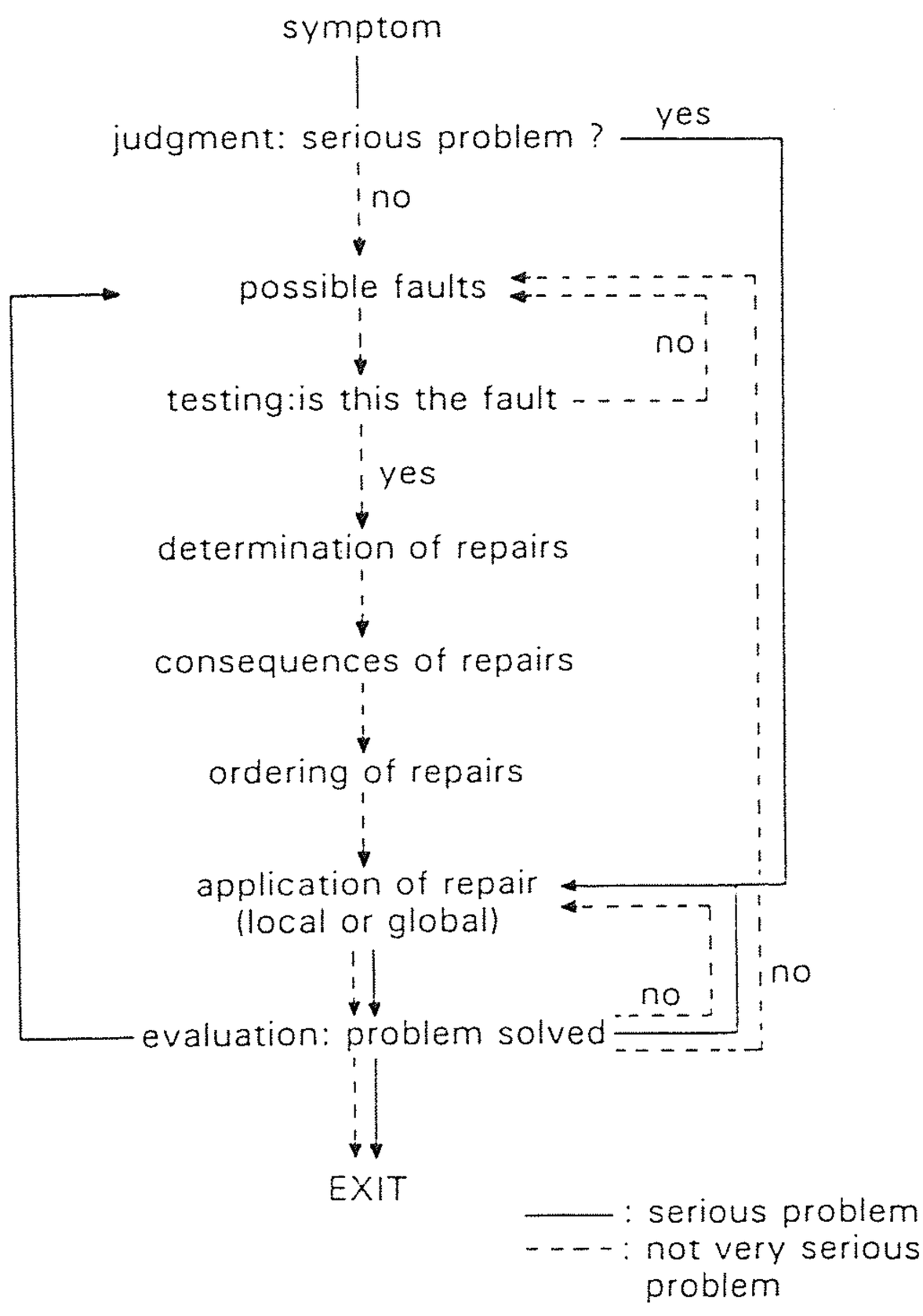

Figure 1. Global strategies as applied by expert operators.

different. Figure 1 gives an overview of the different strategies an expert operator may follow.

If the problem is judged to be serious, he or she will immediately continue with the application of a (global) repair, followed by an evaluation whether the problem has been solved. This process may be followed by a more complete diagnosis in which the fault causing the alarm is diagnosed in order to determine the correct local repair, ensuring a solution 'once and for all'. If the problem is not a very serious one, the subject will consider possible faults one by one and test them, until a likely one is found. This is then followed by a determination of repairs, their consequences, an ordering of repairs (if necessary), application of repairs and an evaluation whether the problem has been solved. If not, the expert might do two things: either try another repair, or back up higher in the tree: he or she may realize that he or she has not yet spotted the actual fault, and therefore the problem has not been solved. In case no possible faults are left, or the operator cannot think of any other faults than the ones he already tested, he or she will be inclined to use a global repair to alleviate the problem.

The model describing the diagnostic strategy employed by relatively inexperienced operators differs from the expert model and is far more simple. Firstly, some of the categories (judgement, evaluation, and consequences) are lacking altogether. Also novices jump much more quickly to repairs, without realizing whether a certain repair actually is right and optimal for a certain situation. Finally, novices appear to 'forget' to evaluate their problem solutions.

Thus, in conclusion, the task structure for diagnosis in technical environments may be qualitatively different for experts and novices in a certain domain. If one wants to use the task structure as an interpretation scheme for the development of knowledge-based systems, the task structure used by expert subjects may serve as the normative task structure, and can be used as such.

4. Diagnosis in technical environments: local strategies

Much of the work that may be interpreted as investigating local strategies in diagnosis has to do with local strategies that are used to determine the fault given the set of symptoms. In terms of the task structure, this fault finding process includes the process of testing, and an ordering of the faults according to likelihood. As described by Jansweijer *et al.* (1989), sometimes symptoms are strong indicators for certain faults and relate the symptom more or less directly to the repair. Often these relations are indirect. In that case, data obtained from observing the installation will have to be abstracted before they can be associated with possible causes for this malfunctioning, or a possible solution has to be refined to an executable remedy. This type of reasoning is known as *heuristic classification*. A slightly different association is the one where a symptom can be translated into disturbed functions. For example, a puddle of water underneath the washing machine probably leads to the conclusion that the water-containing function of the washing machine is disturbed. It is best to focus on that function and to exclude other functions such as the timer-control or motor-function. A third type of knowledge that is used to interpret symptoms is knowledge about the expected appearance of the 'inside' of the device. The physical appearance of parts and components is compared to an expected outlook. A loose wire has to be connected again. Otherwise symptoms like for instance a hot or burnt component, smoke, sparks or a strange sound give an immediate focus for diagnosis. As for all these three types of shallow knowledge used to interpret symptoms into a certain direction, they may be misleading, in a sense that a component may look all right, but in fact is not (such as may happen with plugs).

Quite relevant with respect to search strategies from symptom to underlying fault is the work by Rasmussen and coworkers, as described in Rasmussen (1986). According to Rasmussen (1986), diagnostic search strategies can be performed in basically two ways. A set of observations representing the abnormal state of the system—a set of symptoms—can be used as a search template to find a matching set in a library of symptoms related to different abnormal system conditions. This kind of search is called *symptomatic search.* Symptomatic search consists of a form of pattern-matching between the symptoms and the result in terms of a label, which may be a cause, effect, locations or an appropriate control action. The second type of search strategy as identified by Rasmussen is *topographic search*. A topographic search is performed by a good/bad mapping of the system through which the extent of the potentially 'bad' field is gradually narrowed down until the location of the change is determined with sufficient resolution to allow selection of the appropriate action. Topographic strategies depend on search with reference to a *model of normal function*, and are therefore well suited for identification of disturbances that are not empirically known. Thus, topographic search strategies are not tied to a specific context and are context-free. In contrast, symptomatic strategies can only be applied when a library of symptom to fault connections is at one's disposal. Therefore, symptomatic strategies are tied to specific contexts for their application.

Some of the different local strategies that have been described in the literature may be considered more powerful than others. Obviously, heuristics (or symptomatic search), connecting symptoms to underlying faults, belong to the most powerful strategies an operator may have, although they are only applicable in a narrow domain, and have no wider generality beyond this domain. Therefore, they are likely to fail in any new situations. Less powerful, but still leading to conclusions rather efficiently, are search strategies such as topographic search, geared towards diagnosis in technical domains, but more widely applicable than heuristic search. At the next level of generality are local strategies such as split-half approaches, in which the goal is to minimize the number of tests to localize the source of the failure. At the lowest level of specificity are so-called weak problem-solving methods, such as means-ends analysis and generate and test, as described by Newell and Simon (1972). These search mechanisms are asserted to be a central component of general problem-solving skill and are very general in scope, thus trading power for generality.

The search strategies used by technicians may be different in different situations, and may also depend on the level of expertise. Since heuristics are mostly developed through practical experience and are tied to specific situations, they may become increasingly available with increasing levels of expertise. As demonstrated by Rasmussen (1976), topographic search strategies, although not informationally economic, may be preferred by technicians in domains such as electronic troubleshooting.

Although most of the research on search strategies focuses on fault finding, some research has been done investigating the strategies people use to optimize the number of tests needed to localize the source of the failure. Interesting in this respect are experiments carried out by Morrison and Duncan (1988), in which people were presented with novel failures in a context-free network task. According to these researchers, this task is comparable to electronic circuitry troubleshooting or troubleshooting with respect to powerlines in a telecommunications system. They found that in this task it was more efficient in terms of number of tests if subjects spent more time on utilizing overview information on the system, instead of simply testing units sequentially from right to left. However, it should be noted that the former strategy puts a much higher demand on working memory resources, which are only limited available, and secondly, that the less efficient strategy did not necessarily lead to a lower diagnostic performance. Schraagen and Schaafstal (1991) in a study on electronics troubleshooting (radar mechanics) showed that a 'split-half' strategy was a rather efficient test strategy, once the area in which the cause of the problem could occur was reduced to a manageable size.

5. Diagnosis in technical environments: underlying domain knowledge

Now that we have addressed what has been found about the task structure in diagnosis and the corresponding local strategies, the question remains what kind of domain knowledge plays a role in diagnosis in technical environments. Early work on expert systems seemed to imply that there was only one underlying model of domain knowledge. Work along this line gave rise to the *model-based approach*, which postulates that expert system building should start with an encoding of the first principles of a domain, for example, qualitative or quantitative models of the behaviour of the device to be diagnosed (e.g., Davis 1984). In most of the cases, this work concentrates on models about the structure and behaviour of the device. However, as described in Steels (1990), it is possible to think of a variety of models, each focusing on different aspects of the problem domain. For diagnosis, for

example, one could think of a structural model describing part–whole relationships between components and subsystems, a causal model representing the cause–effect relationships between properties of components, a functional or behavioural model representing how the function of the whole follows from the function of the parts, a fault model representing possible faults and components for each function that might be responsible for the fault, and an associational model relating observed properties with states of the system. Which of these models would be the one to use? Clearly, all these models are useful. Simmons (1988), cited in Steels (1990), for example, describes a system that translates causal models into associational models and shows how they have complementary utility in problem solving. Thus, the question is not what model to use exclusively, but what type of model or type of domain knowledge is appropriate in certain stages of the diagnostic reasoning process.

Most of the research on types of domain knowledge used concentrates on the process of fault finding. Jansweijer *et al.* (1989) for example, found that the type of model that is of primary use to the service engineer of the UV-recorder is a model of the *function* of the device on different levels of abstraction. The service engineer knows the set of subfunctions that realizes a higher level function. The subfunctions can either be a set of independent subfunctions or a series of subfunctions that enable each other. Each subfunction can itself be described in the same way until, at the lowest level, a function is realized by a component that cannot be decomposed further, or of which the decomposition is of no interest since it is not repairable, but has to be replaced entirely.

In this way, a multi-fold representation of the device on different levels of abstraction exists. On the highest level functions are described with global parameters covering functions as expected by a naive user. At the lower level, the functions are described with parameters covering more detailed and inner functions of the device. A high level function of a washing machine for example is 'the cleaning of dirty wash', with a parameter of the 'cleanliness' of the linen. A lower level function of a washing machine for example would be heating the water.

The knowledge about the domain that the service engineer exploits is sometimes of a generic type. On a high level, knowledge about the function is device independent and independent of the physical realization. A service engineer that knows to locate and replace, for instance, a defective water-pump in a washing machine, knows to do the same in a washing machine of a different brand. Moreover, he or she is presumably able to locate and repair the same component in a machine with a somewhat different high-level function, such as a dish-washer. Having knowledge about high-level generic functions is an efficient way for representing knowledge about systems since it can be used in other, different domains.

To summarize: the multi-fold representation of the device, on different levels of abstraction has a hierarchical structure. Higher functions are realized by the sub-functions it subsumes. High levels describe functions independent of the physical realization. On the lowest level, the model corresponds to a static description of all the components and the relations between these components. On this level the service engineer usually has to depend on service manuals and schemata. But this level of representation is usually too detailed for the technician to be of any use and seldom needed as well. In standard repair practice, complete assemblies of components will be replaced rather than the individual components.

If we take a more detailed look at the types of knowledge people have about devices, it is clear that people know a lot more about the device than just the

functioning and the structure of a device. For example, Kieras (1982, 1987) showed that people have many types of knowledge available, such as the name of the device, its function, the controls and indicators, the connections with other devices, the power source and requirements, the internal and external layout, procedures for operating and troubleshooting the device, and knowledge about the internal structure and mechanisms. Kieras (1982, 1987) suggests that most knowledge about devices is related to *using* the device, as opposed to *how-it-works knowledge* about the internal structure and operation of the device. For example, Kieras (1982) observed that when asked to freely describe a device, experts would provide considerable detail on the procedure for using a piece of equipment, but often they did not consider it necessary to provide any details about how the device worked, although it was very clear that they would be able to do so when asked. The question is, though, whether the types of knowledge people use and report depends on their job-contents (an operator vs. a technician, or a technician vs. a manager), and there is suggestive evidence in the literature that this is in fact the case (Cuney 1979, Rasmussen and Lind 1981, Schaafstal 1989).

If we take a closer look at the relationship between various types of underlying domain knowledge and stages in the diagnostic process as described in section 2, the following relationship is suggested by Schaafstal (1991), taken from an industrial domain (the paper and pulp industry):

Table 1. Relationship between phases of the model and used system knowledge. 1 = symptoms, 2 = judgement, 3 = possible faults, 4 = ordering of faults, 5 = testing, 6 = determination of repairs, 7 = consequences of application of repairs, 8 = evaluation.

	1	2	3	4	5	6	7	8
Process flow		*	*		*		*	
Top. location						*		
Controls	*		*		*	*	*	
Function comp.	*	*	*			*	*	
Paper making	*	*	*	*		*	*	*
Normal values	*		*		*			*
Process dynamics	*		*		*			*
Functioning comp.			*			*		

However, much more research in different domains is needed to establish more firmly this kind of interaction between types of domain knowledge and stages in the diagnostic reasoning process.

This section may be concluded by the following remarks. Much of the earlier work on model-based reasoning focused on models of the structure and behaviour of the device. Although this work has led to very interesting insights, in the long run this seems to be a too restrictive viewpoint, as pointed out by Steels (1990). Thus, a more flexible view towards the use of domain knowledge is needed, certainly if we take into account the flexibility with which people use different types of knowledge, at different levels of abstraction and presumably also at different stages of the diagnostic reasoning process.

6. Implications for the training of diagnostic skill

One of the most striking findings in the literature, as summarized in Morris and Rouse (1985) is that instruction in theoretical principles is not an effective way to produce

good troubleshooters. It is interesting to note that these results are quite consistent with reports from other domains such as process control (Brigham and Laios 1975, Crossman and Cooke 1974, Kragt and Landeweerd 1974, Morris and Rouse 1985, Mayer and Greeno 1972, Mayer *et al.* 1975), in which explicit training in theories, fundamentals, or principles failed to enhance performance, and sometimes actually degraded performance.

However, when theoretical instruction is combined with training people in how to use that knowledge, performance usually gets better. For example, Miller (1975) described an experimental training course for radar mechanics, in which an effort was made to relate instruction in system functioning to actions performed during troubleshooting. For example, system behaviour was presented in terms of causal sequences rather than a more traditional presentation of schematics. Whenever possible, attempts were made to relate schematics to the actual equipment. A control group received instruction in the theory upon which the system was based and a left-to-right presentation of schematics, with no extraordinary attempt to relate the information to the actual equipment. Both groups had limited access to the actual radar for troubleshooting practice. The experimental group turned out to be faster in performing checks and adjustments, was more often successful in troubleshooting, made fewer errors in general, and scored better on quizzes.

It should be noted, though, that in the studies in which positive effects were found, the guidance involved was rather explicit: students were told to generate hypotheses, chunk information, and analyse symptoms in a prescribed way. This is a far more active approach than just providing an opportunity to use system knowledge, and should not be interpreted as evidence that the latter approach will produce better troubleshooters.

Thus, one of the elements that helps in training people to become better troubleshooters is an explicit guidance in the use of previously acquired theoretical knowledge. Another important factor in training of troubleshooting is the opportunity for practice (e.g., Johnson and Rouse 1982a, 1982b).

Not much is known about training in local strategies. A number of studies showed that supplying people with adequate procedures can have a positive effect on their troubleshooting performance (Potter and Thomas 1976, Smillie and Porta 1981, Elliott 1966). There is also limited evidence available that providing troubleshooters with good examples can have a beneficial effect on their performance (e.g., Johnson and Rouse 1982a, 1982b). However, learning from examples may be confined to people with high ability, and appears to be dependent on explicit instruction to learn from the examples as well.

Training with respect to task structures is still a rather neglected area, although the importance of strategic control knowledge in the form of control of goal structures has been shown in a number of domains (Greeno 1978, Card *et al.* 1983, Anderson *et al.* 1987, Kieras 1987, Gott and Pokorny 1987). It happens that in instructional materials a number of time gaps have been identified with respect to goal structures (Greeno 1978, Anderson *et al.* 1984, Kieras 1987). The emphasis on the importance of training people how to use strategic knowledge is also one of the most important outcomes of the Guidon project (Clancey 1987). Training of explicit goal structures is not impossible, though, since it has already been accomplished a number of times, for example in some of the intelligent tutoring systems, such as Proust (Soloway and Johnson 1984), and the Lisp tutor (Anderson and Reiser 1985), but this area still needs a lot of research.

This section may be concluded with the recommendation of Morris and Rouse (1985) that the most promising instructional approach worthy of further research is one where procedures for approaching fault isolation problems are combined with instruction in how to use system knowledge in deciding exactly what to do.

7. Conclusions

The framework as described in this paper offers the possibility to interpret results obtained from various sources about diagnostic skill. It also enables a more thorough comparison between diagnostic tasks in different domains and opens a number of research questions. For example, what is the interaction between local strategies and underlying domain knowledge? A second research question concerns the applicability of certain search strategies in certain domains. Will it eventually be possible, based on a characterization of the domain, to predict which strategies will lead to good performance? From a more applied viewpoint, the question whether it is possible to train effective strategies is still unanswered. Based on an investigation of various diagnostic tasks and a comparison between them we are currently working towards answers with respect to these questions.

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